Noice Cancellation-Normalized LMS Algorithm

第10組
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Normalized LMS Algorithm

$$y(n) = w^T(n)x(n) = \sum_{i=0}^{M-1} w_i(n) \bullet x(n-i)$$

$$e(n) = d(n) - y(n)$$

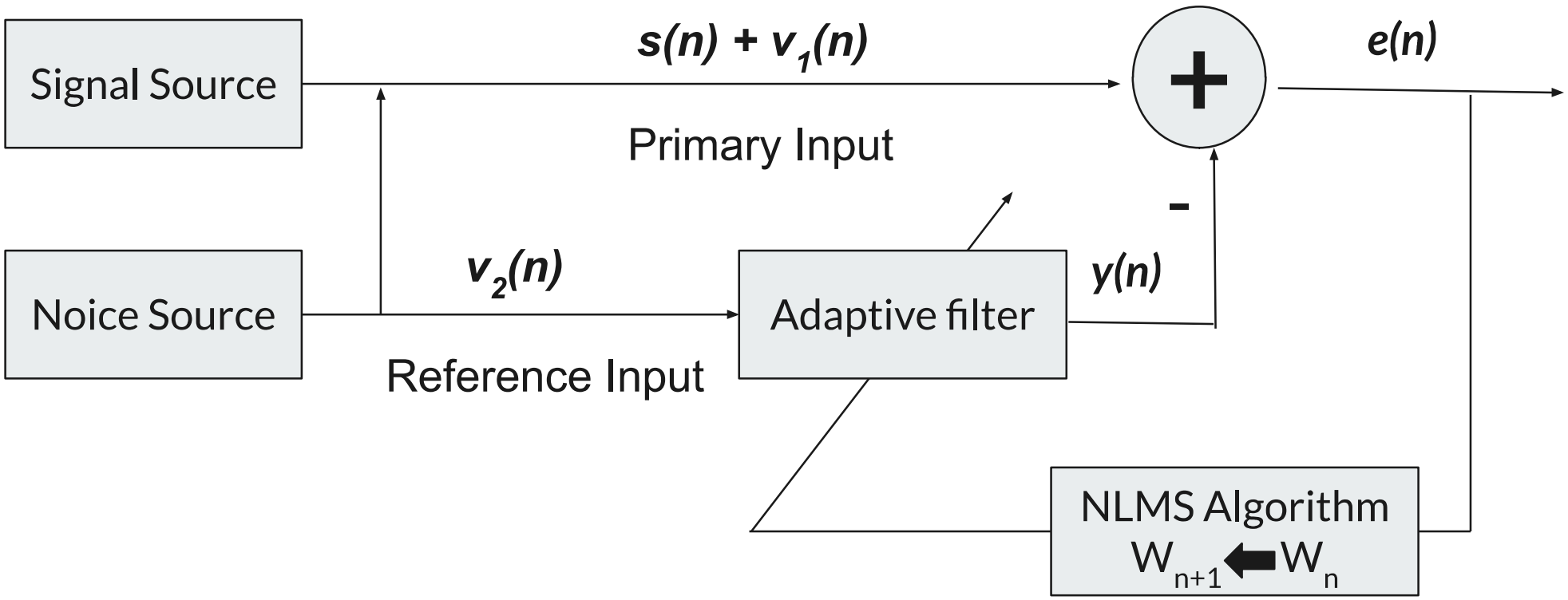
$$w(n+1) = w(n) + \frac{\mu}{\varepsilon + ||x(n)||^2} \bullet e(n) \bullet x(n)$$

$$w_i(n+1) = w_i(n) + \frac{\mu}{\varepsilon + ||x(n)||^2} \bullet e(n) \bullet x(n-i)$$

where i = 0,1,2,...,M-1

M:濾波器階數
μ:步長參數
x(n):輸入訊號
y(n):輸出訊號
d(n):理想輸出訊號
ϵ:避免除以零的小常數

Noice Cancellation



Noice Cancellation

$$y(n) = w^T(n)v_2(n) = \sum_{i=0}^{M-1} w_i(n) \bullet v_2(n-i)$$

$$e(n) = d(n) - y(n)$$

$$w(n+1) = w(n) + \frac{\mu}{\varepsilon + \left\|v_2(n)\right\|^2} \bullet e(n) \bullet v_2(n)$$

$$w_i(n+1) = w_i(n) + \frac{\mu}{\varepsilon + \left\|v_2(n)\right\|^2} \bullet e(n) \bullet v_2((n-i)$$

where i = 0,1,2,...,M-1

Noice Cancellation

利用(e(n)=訊號源−y(n))剩餘的訊號，即希望保留的語音

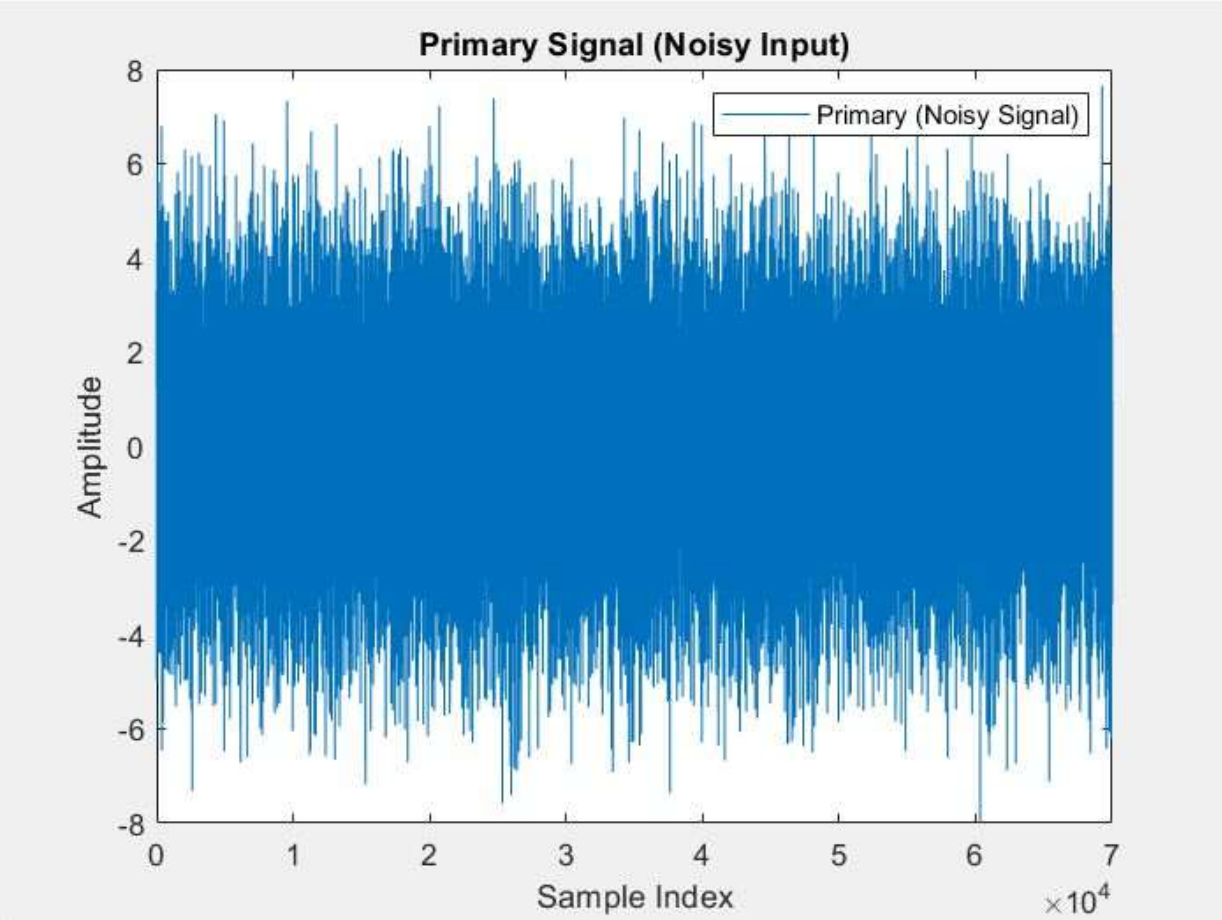
參數調整

- 濾波器階數(Filter Order)：影響模型的學習能力，階數高可擬合更複雜噪音
- 步長參數μ：太大會發散，太小則收斂慢

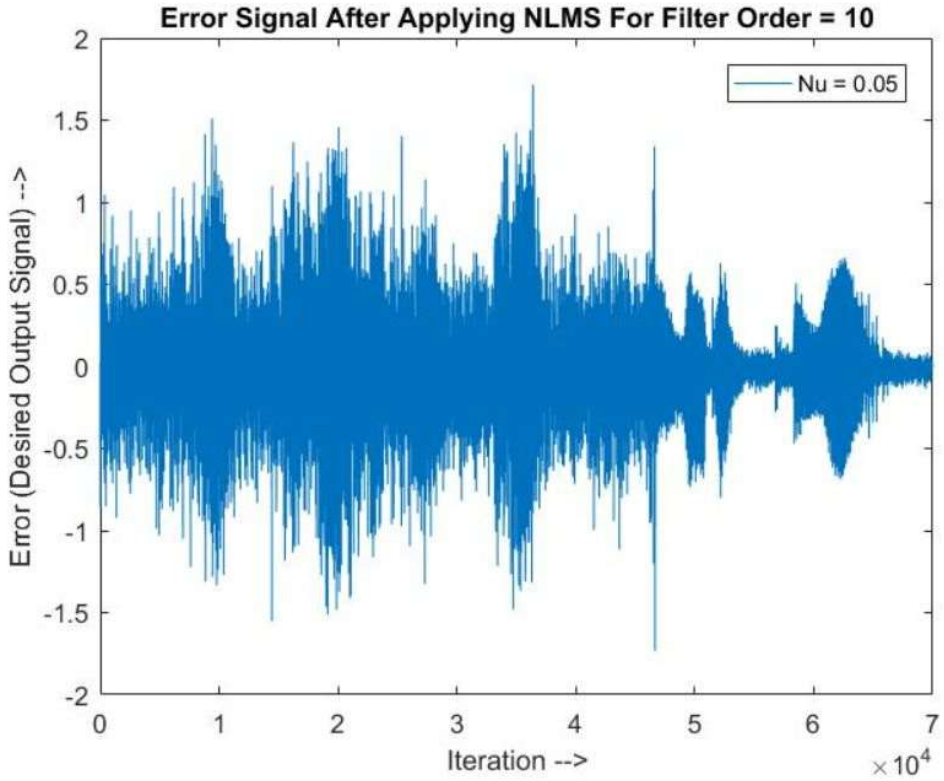
目的: **濾波器會在多次迭代後收斂，也就是誤差 e(n) 穩定下來，這時可以視為已成功去除大部分噪聲**

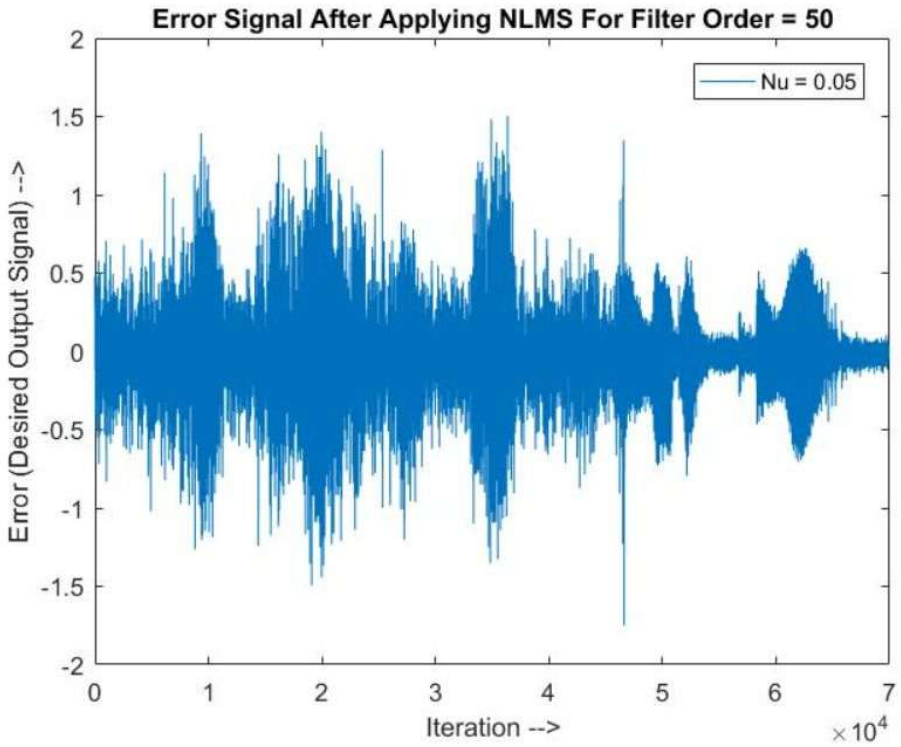
模擬結果

Primary Input



模擬結果


Error Signal After Applying NLMS For Filter Order = 10
Nu = 0.05
Error (Desired Output Signal) -->
Iteration -->
×10^4
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
1
2
3
4
5
6
7


Error Signal After Applying NLMS For Filter Order = 50
Nu = 0.05
Error (Desired Output Signal) -->
Iteration -->
×10^4
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
1
2
3
4
5
6
7

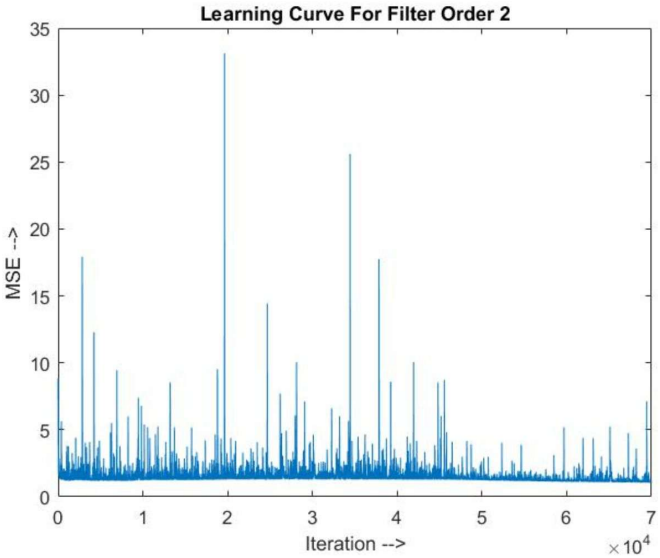
實驗模擬結果

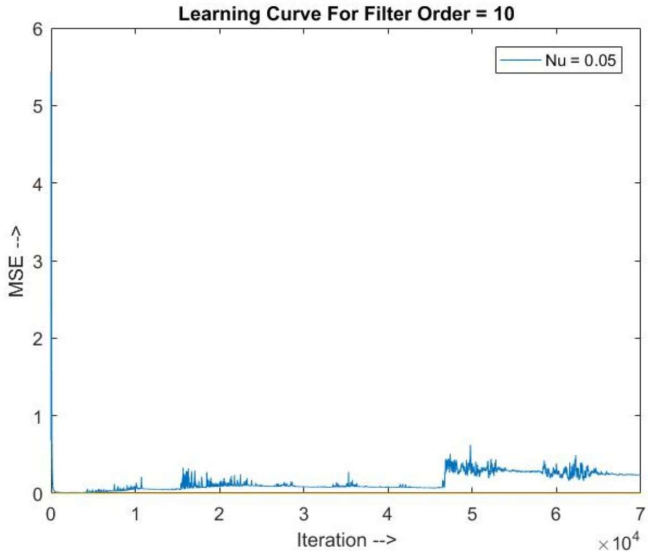
原始聲音(人聲+噪聲):

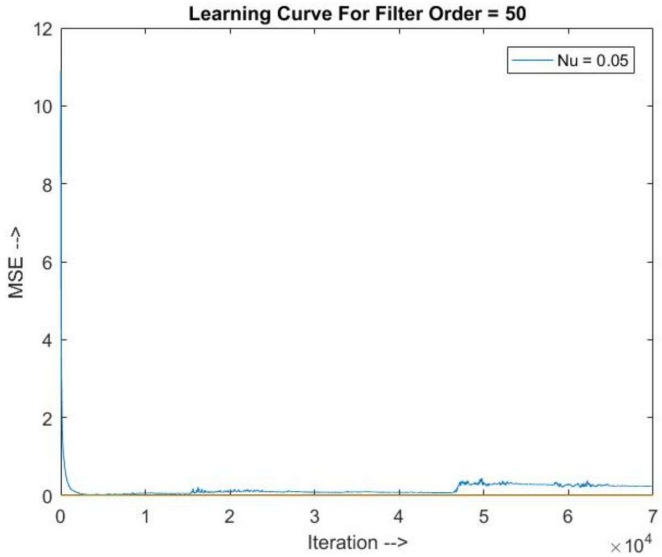
訓練聲音(噪聲):

濾波器輸出結果 :

Filter Order (M)多寡對MSE造成的影響







M上升時

優點: 較能夠抵抗雜訊,更好的還原訊號，並收斂速度加快。

缺點: 計算複雜度上升。

改變步常參數（μ)

若將μ設太大:權重更新幅度也相對加大 ,收斂速度較快, 但容易震盪使得難以穩定收斂一個 值。

若將μ設太小:導致收斂速度緩慢

實驗:

不同μ值的MSE比較、不同μ值對Tap Weight 造成的影響

預期結果 :

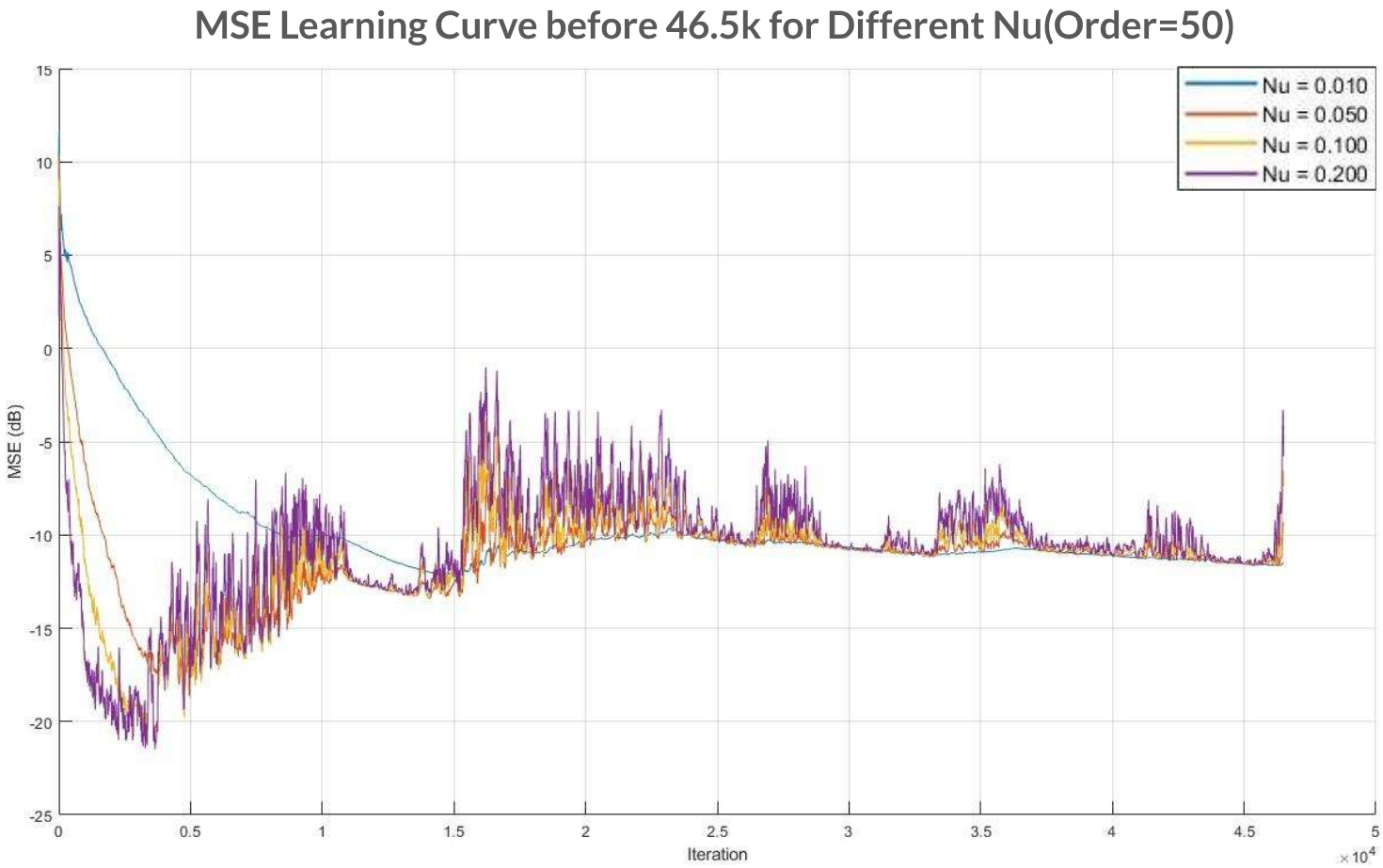
μ 值= 0.01 | 穩定但收斂慢

0.05 | 通常為最佳平衡

0.1 | 收斂快但可能有震盪

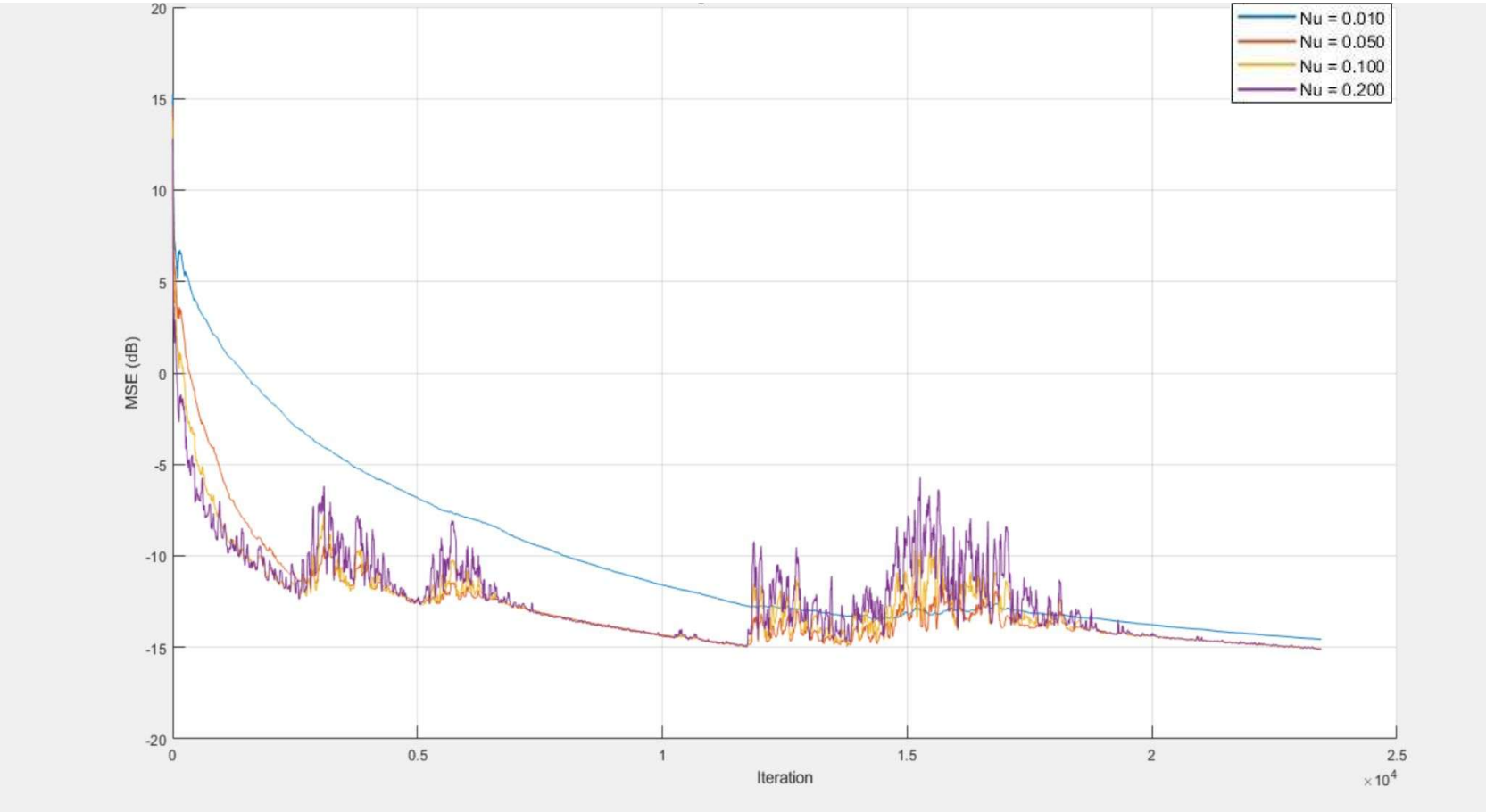
0.2 | 有機會發散或過擬合

不同μ值的MSE比較


MSE Learning Curve before 46.5k for Different Nu(Order=50)
Nu = 0.010
Nu = 0.050
Nu = 0.100
Nu = 0.200
MSE (dB)
Iteration
×10^4
15
10
5
0
-5
-10
-15
-20
-25
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

不同μ值的MSE比較

MSE Learning Curve after 46.5k for Different Nu(Order=50)



步長參數μ對tap weight所造成的影響

發現:

M=1 : 僅管μ設定小值，也會導致tap weight更新權重過快,使得難以收斂

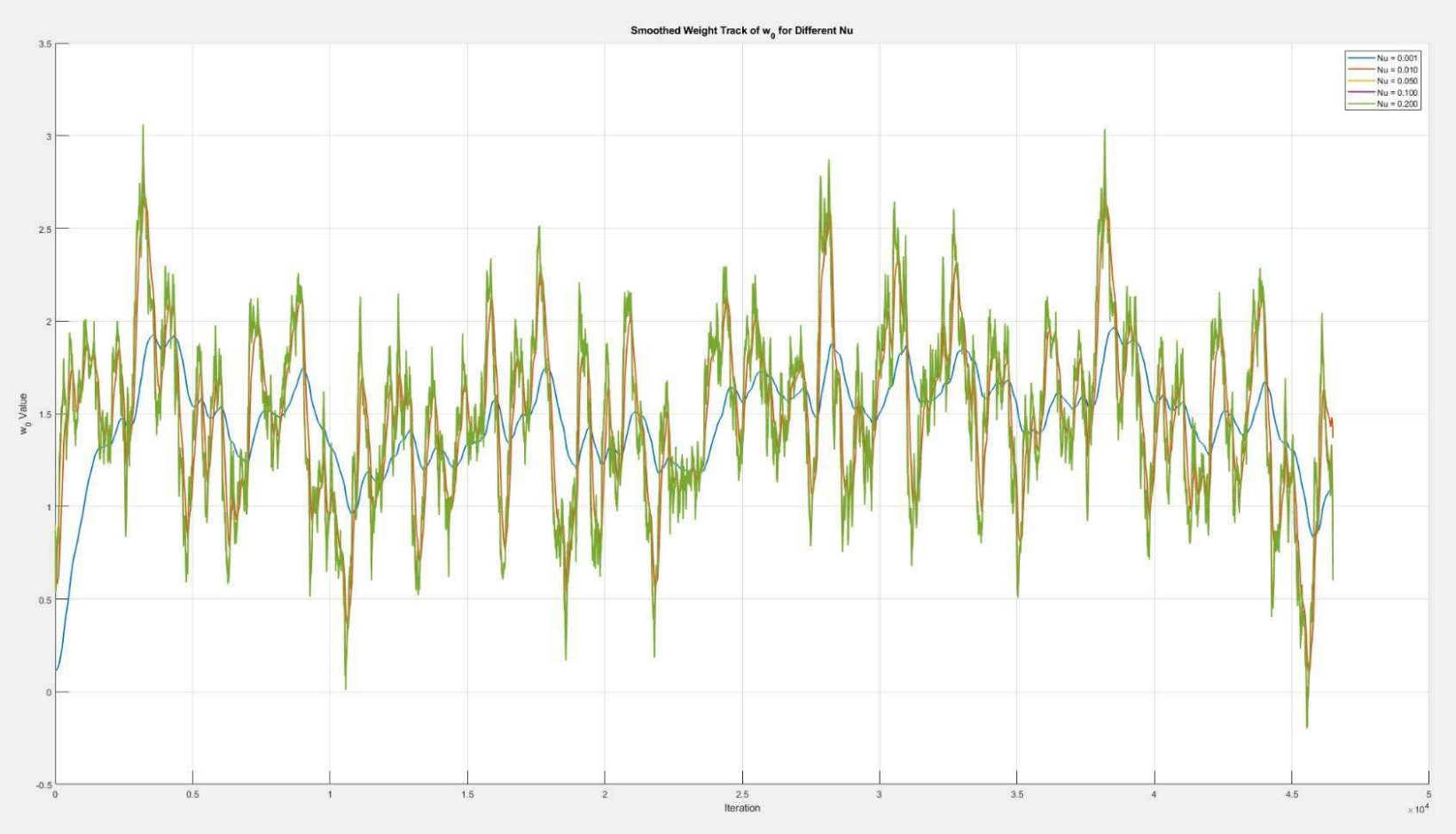
M=50 : 恢復正常狀態，便可觀察μ越大，收斂速度越快且易震盪的情形

結論:

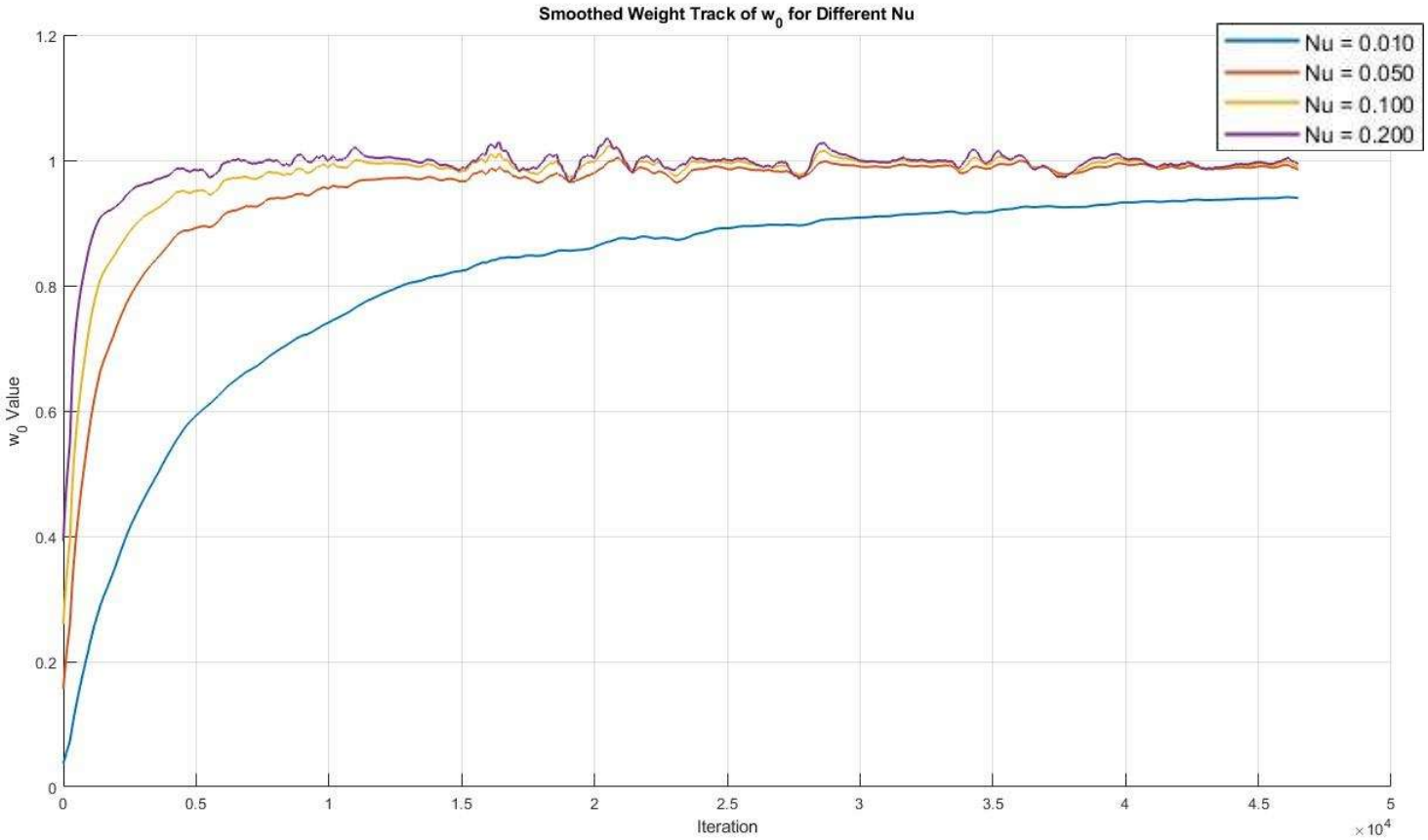
若將μ設太大，則收斂速度較快，但容易震盪，較難以收斂穩定一個值。

若將μ設太小，則收斂速度緩慢

With M=1， μ= 0.01、0.05、0.1、0.2



With M=50, μ= 0.01、0.05、0.1、0.2



參考資料

[1]Mewara, M. (n.d.). *Normalized LMS Algorithm – Interference Canceling* [PDF].

[2]**"Least Mean Squares filter"**, Wikipedia,

https://en.wikipedia.org/wiki/Least_mean_squares_filter

[3]**Dr. R. D. Kharadkar**, *"Acoustic Noise Cancellation using Adaptive Filters: A Survey"*,

International Conference on Electronic Systems, Signal Processing and Computing Technologies, 2014.

[4]**Pramod R. Bokde, Nitin K. Choudhari**, *"Implementation of Adaptive filtering algorithms for removal of Noise from ECG signal"*,Choudhari Pramod R Bokde et al,*Int. J. Computer Technology & Applications*, Vol 6 (1), pp. 51-56.

[5]**Mohd Zaizu Ilyas Ali, O. Abid Noor, Khairul Anuar Ishak, Aini Hussain, Salina Abdul Samad**,

"Normalized Least Mean Square Adaptive Noise Cancellation Filtering for Speaker Verification in Noisy Environments",

International Conference on Electronic Design, 2008.